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HALO REACH: Project Coordination, Market Analysis and Business & Financial Strategy

UAS for Disaster Relief - AE3 2024

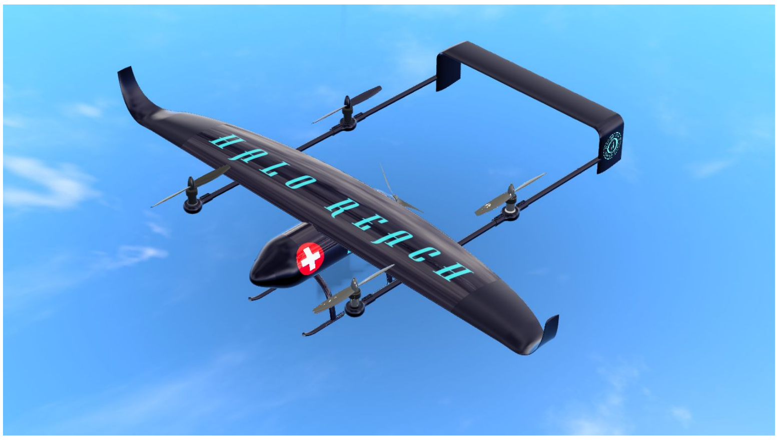

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Table 1. People involved in HALO REACH - GDP (**student in bold**)

Team code	Team	Names	CID
FATT01	**Team Leads**	**Moreno, Patricia**	**01857449**
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		Dirirsa, Naboth	02026211
		Lawlor, Max F	02026199
		Singh, Parasdeep	02099678
		Homa, Matthew M	02038403
FATT03	Structures	Leng, Jiaxun (Aaron)	02015060
		Wang, Zhiyan	01911209
		Liu, Stephen	02015258
		Chauhan, Shivani	02096006
FATT04	Systems and propulsion	Lin, Yongkai	02029838
		Chursuk, Tharm	02048996
		Wankhade, Divij P	02027906
		Aggarwal, Ishaan K	02030969
		Halim, George	02059571
FATT05	Flight dynamics	Baikady, Thilak S	02016794
		Scott-Brown, Ruairidh	01734080
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		Tang, Shun S	02023280
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Abstract

This report presents the business design of HALO's Uncrewed Aerial Vehicle (UAV): REACH . A versatile UAV with a 160 kg Gross Takeoff Weight (GTOW) engineered for disaster relief. REACH offers 10+ hour endurance missions carrying 20 kg payloads, or 20-minute missions with up to 50 kg payloads. The UAV's modular systems enable applications in agriculture, surveillance, large-scale delivery missions, and infrastructure management, making it a 5-in-1 versatile solution.

Our marketing strategy targets the growing infrastructure and agriculture market segments, focusing on their need for automation and efficiency. This approach aims to secure profitability and funding for the business while assisting private companies, governments, and unions in disaster mitigation - our primary mission. The report covers the UAV's life cycle phases, case studies in agriculture and disaster relief, and the dynamic reorganisation of team roles, crucial for the HALO REACH's success. Finally, the report shows a detailed financial analysis over a ten-year forecast - until the company is expected to reach maturity -, identifying a break-even point in the 5th year and an initial funding requirement of $18 million.

Overall, HALO REACH represents an attractive investment opportunity in a startup that leverages cutting-edge technology in a versatile multi-use UAV with a competitive pricing strategy and a commitment to ESG principles.

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Nomenclature

DoF	Degrees of Freedom
ESG	Environmental, Social, and Governance
GTOW	Gross Takeoff Weight
HALO	Humanitarian Aid Logistics Operations - the company name
M	Million
REACH	Rapid Emergency Aid and Communications Hub - the company's UAV name
UAS	Uncrewed Aerial System
UAV	Uncrewed Aerial Vehicle
VTOL	Vertical Take-Off and Landing

1 Introduction

Natural disasters have increased almost five-fold in the last 50 years and economic losses in the affected regions have increased tenfold [1]. Following these disasters, conventional routes to disaster zones become inaccessible to various vehicles due to flooding, road damage, or runway destruction. In such scenarios, UAVs, specifically VTOL UAVs, offer a great advantage to the conventional automotive and aircraft. With current and near-future technology and the outlook in UAV market, society is aiming to create disaster relief UAVs the norm. The proposed project outlines the design and business considerations for a new UAV, REACH, aimed at addressing these challenges and ensuring a profitable product launch.

One key feature of the REACH is its versatility from modular add-on subsystems, tackling disaster relief missions but also offering solutions for agriculture, land monitoring, infrastructure management, and large-scale delivery services. This multi-functionality diversification ensures the profitability of the drone while supporting a more efficient global environment and aligns perfectly with the growing popularity of ESG (Environmental, Social, and Governance) products and technology.

1.1 Aircraft's disaster relief mission requirements

Below we can find a summary of the UAV project requirements to better understand the constraints to our design, where the backslash is used to separate the long endurance to the supply delivery missions. A more detailed discussion of the mission can be found in my co-coordinator's report [2], which ties these requirements to the conceptual design of our aircraft.

- **GTOW**: 160 kg
- **Transmission Radius**: 185 km in 1.25 hours
- **Endurance**: 10 hours / 20 minutes
- **Payload Capacity**: 20 kg / 50 kg in 0.05m^3
- **Propulsion System**: Hybrid
- **VTOL**: 6x6 m^2 platform from a ship's deck
- **Modular Add-ons**: Swappable by two individuals within 10 minutes and $\leq$ 15% of GTOW

The primary mission of this UAV is to take off from a ship, travel 185 km, loiter for either 10 hours or 20 minutes, deliver a payload of either 20 kg or 50 kg, and then return 185 km to land back.

2 Team organisation & Communication

2.1 Information Management

Ensuring effective communication and information management was crucial for the success of the project. Prior to the official start of the project, I ensured that all members were added to a WhatsApp community, a Microsoft Teams channel and a share OneNote. The WhatsApp community included channels for FATT01 announcements, a general chat, and sub-team chats. WhatsApp was used for daily urgent messages due to its popularity and engagement among team members, Teams was used for formal announcements and messages requiring broader visibility and OneNote was used for collaborative written documents. All supervisors were added to Teams to ensure direct and easy communication. Throughout the project, three key files were available to all in the general chat:

1. Weekly presentation slides
2. LaTex report template
3. General information spreadsheet

The first two are self-explanatory, seen in Appendix A. The general information spreadsheet contained several useful sheets for all students, including Gantt charts, easy access to everyone's contacts, sub-teams and emails, weekly presentation assignments, and a shared parameters spreadsheet. I ensured this spreadsheet was available from the first week to all members, assigning responsibility for each parameter to a specific team and allowing each team to add more parameters to themselves or to other teams when required. This allowed other teams to check the most updated values for parameters in real-time, facilitating a smooth integration between teams. Teams could automate their code to read directly from this spreadsheet and run their code with the most updated version of the UAV.

2.2 Feedback & Coordination Strategy

Initially, we implemented various feedback forms to monitor our progress throughout the project:

1. Daily feedback form
2. Post-presentation feedback form

The former was used by the students to provide us coordinators with a more in-depth view of their daily activities and to reflectively identify issues they were addressing, offering insights that might not emerge in person. The latter was used by the supervisors to give the entire team general written feedback on the weekly presentations. Both forms can be seen in Appendix B.

In regards to the first one, us coordinators realised that team members were reluctant to complete the daily feedback forms as they required extra time and lacked motivation. Therefore, I took it upon myself to hold meetings with each group. We met daily to discuss prioritisation and workflow and if needed I scheduled some time for the team and myself to meet and discuss the progress and future deadlines more in depth. For instance, during the second week, we faced significant constraints due to the unfinished weight and balance analysis as FATT03 and FATT05 needed them urgently to continue their part of the project. It was thanks to those slotted meetings that I could gain more in depth view of how this analysis affected the sub teams. Consequently, I asked FATT04 to prioritise this task. In addition to these meetings, I attended sub team's meetings with their supervisors to closely monitor progress and feedback during the first weeks, as these were more organisation based and not as focused on technical results yet, ensuring better coordination across all groups. Conclusively, we adopted a more relaxed approach based on group preferences and avoided requesting extensive written feedback. This day-to-day approach, however, was supported and facilitated by global and sub-team Gantt charts, which will be explained in detail in Section 2.2.1.

On the other hand, the second form was used for post-presentation feedback. Initially, it allowed all supervisors to provide general feedback to the entire group after weekly presentations. However, in the second to third week, it was modified to deliver feedback to individual sub-teams, based on our supervisor's suggestion. This introduced more individualised and useful written feedback, in addition to the oral feedback during class, which I summarised and documented in OneNote.

2.2.1 Gantt Charts

As explained, the relaxed approach adopted by the team needed close monitoring of overall progress and internal deadlines set by FATT01. Therefore, we implemented Gantt charts, found in Appendix C, accessible to all team members. These charts ensured that everyone was aware of their current and future tasks, as well as dependencies on other teams. Some benefits of this model include:

- **Progress Tracking:** Sub-team Gantt charts enabled us to monitor individual teams' progress.
- **Identifying Productivity Gaps:** Daily meetings and Gantt charts helped identify and address productivity gaps early.

One such productivity gap involved handling an individual's lack of motivation, causing another team member to compensate for the shortfall. My co-coordinator and I noticed the work discrepancy and addressed it through group and individual discussions. The aim was to resolve the matter without escalating it further. Finally, the individual returned to their expected performance level.

2.2.2 Weekly presentations organisation

Weekly presentations were organised by FATT01, and I took the lead in coordinating them. In the first week, I consulted with the supervisors to find a day that would suit everyone best for these meetings, settling on Fridays from 10-12 am. If the date had to be changed, I used Doodle amongst the supervisors to obtain the best date for everyone and ensure maximum attendance.

As per requirements, every team member had to present at least twice during the 5-week period. Therefore, I created an Excel sheet with a counter, automatically checking how many times one had presented and ensuring even collaboration amongst the teams. Run-throughs were conducted the day before, where FATT01 encouraged Q&As to ensure everyone provided feedback to each other.

Based on supervisors' feedback, I created a uniform template for cohesive presentations. At the start of each presentation, I briefly reintroduced our UAV and its main mission to ensure all supervisors were on the same page, as one can forget details from week to week. Clarity was emphasised with minimal text and visually appealing slides, and animations were introduced, with the latter being a valuable addition from FATT04. The presentation order changed each week to prioritise groups with the most significant breakthroughs and not penalise the last ones in case the group was time constrained. This became an issue towards weeks 2 and 3, so I ensured that during the run-throughs, each member timed themselves speaking to stay within a 12-15 minute timeframe per team to allow question/feedback time. This got progressively better, although it was always a constraint for the entire group, and more emphasis should have been put on it.

3 Market analysis

Understanding the market is critical to any business to succeed and ensure its long term profitability. As such, we conducted a comprehensive market analysis, drawing key information from analysts like Droneii, MarketsAndMarkets, Mordor Intelligence and Fortune Business Insights. We also consulted analyses from advisory companies such as PricewaterhouseCoopers (PwC) and Scyne Advisory, in addition to news and articles from sources like DroneLife, among others. We structured our market analyses in three sections following the three big questions posed by McKinsey&Company [3]:

3.1 What market segments offer higher growth potential and attract investors?

Table 2 provides a summary of our market analysis, offering a brief overview before delving into a detailed discussion:

Table 2. Comparison of Market Segments Summary

Segment	Advantages	Limitations	Decision
Military Missions	High profitability	Deviates from core values, impacts investor confidence	Not Pursued
Last-Mile Delivery	High demand	Price competition, small UAV focus	Not Pursued
Infrastructure	Improves safety, project management, reduces costs & time	Initial investment in technology and training	Pursued
Agriculture	Large-scale drone capabilities, reduces pesticide usage, environmental benefits	Market adaptation required, competition with existing agricultural drones	Pursued

Our analysis show that the UAV industry appears to be growing at a steady rate, with a 10-16% Compound Annual Growth Rate (CAGR) projected by 2040 [4, 5]. While the UAV market encompasses the range of disciplines illustrated in Fig. 3, this market surge is predominantly driven by two segments: military and delivery UAVs. Military presents one of the most profitable areas and with highest growth capacity, with an expected CAGR of 7-13% by 2028 [6, 7]. The profitability of this segment, clearly influenced by geopolitics, can be attributed to two key factors:

1. UAVs provide more comprehensive data than humans can achieve
2. UAVs do not put personnel at risk

Based solely on the above growth rates, we considered entering these segments by expanding our focus from disaster relief to military missions. However, I suggested that this expansion could deviate HALO from our core social benefit values and compromise investor confidence as certain investment funds are not allowed to invest in a startup with military involvement. Given this paradigm, FATT01 decided to hold a meeting amongst all team members, present our key market findings and seek feedback for our market strategy. It was collectively decided that we would not pursue the military and defence segments and would instead focus on developing an ESG-friendly product.

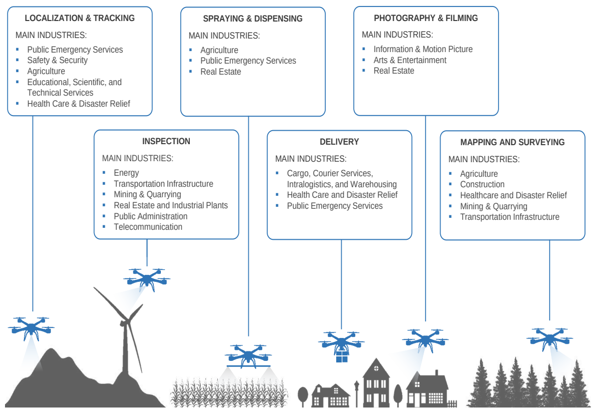

Figure 3. Drone application methods [8]

Following this decision, the next market areas with significant growth potential were the last-mile large-scale delivery, infrastructure and agriculture segments.

Last-mile UAV delivery missions were the most in demand [9], however, this market is predominantly influenced by small UAVs capable of navigating urban environments [10], where price rather than technology is the competitive factor for success. Our UAV design differs greatly from those in delivery missions; HALO REACH is a large-scale drone, designed for efficient 10-hour loitering via a 5.55 m span wing. Moreover, we had established since the outset that our UAV would include cutting edge technology which is not required in the distribution segments. Given these differences, we opted to focus on the remaining two growing areas: infrastructure and agriculture.

In the infrastructure sector, UAVs can significantly improve workers safety and project management. Fixed-wing VTOL drones can perform hazardous inspections in inaccessible areas (e.g. dense vegetation and tall infrastructure), reducing risks for human workers and providing real-time site information to aid in emergency response and evacuation planning. Data from developing countries shows that construction risks are three to six times higher than in other industries [11], and in the US, construction accounts for over 20% of work fatalities despite comprising only 5% of the workforce [12]. Implementing UAVs could substantially reduce these safety risks. For infrastructure project management, fixed-wing VTOL drones can collect data to create accurate 3D models, volumetric calculations, and orthophotos, facilitating planning and decision-making. All this translates into cost savings and reduced timelines. PwC and IDB LAB report that 6 to 7 out of 10 respondents in Latin America found that drones shortened project schedules and reduced costs by at least 10% [13]. Currently, 28% of infrastructure projects suffer from overruns due to various issues such as design errors, site conditions, poor risk management, and logistics disruptions [13]. In summary, UAVs can address labor sub-productivity in construction, which has only increased by 1% annually over the last two decades compared to a global economic growth rate of 3.1% [14]. Conclusively, we opted to enter this market.

The agricultural segment was the next one that offered attractive growth potential. Companies like Hylos in the US, ACUARIS in the EU, and DJI in China make extensive use of drones dedicated to crop spraying and monitoring activities. However, these agricultural drones typically have limited payload capacities and endurance of up to 6 hours. HALO REACH aims to differentiate by offering a larger-scale drone with modular add-on systems for multiple functions: crop spraying with up to 50 kg of pesticides, land monitoring and material supply. Our versatility is very attractive for large-scale farms that require tailored solutions beyond standard crop-spraying drones. Moreover, our drone could reduce pesticide usage by up to 90% by accurately targeting pest infestations which responds to our environmental concerns [15], which minimises the environmental impact and protects beneficial insects. The fact that REACH enables efficient land monitoring, crop spraying and infrastructure monitoring is particularly advantageous in organic farming where strict regulations limit synthetic chemical use [15]. We therefore concluded that the needs for tailor made technology in large scale farms and the environmental requirements of certain farms fit with our offer of cutting edge technology and ESG services.

In summary, the team decided to target the market segments found in Table 3. Going back to the project, I maintained close communication with FATT06, the Modular Add-on Sub-systems Design Team during the whole market analysis process due to their tight relationship with the target markets. We also determined collaboratively which modular add-on systems would be most beneficial for HALO.

Table 3. Targeted mission categories and descriptions

Disaster Relief	Aforementioned mission with an aluminium $0.05m^3$ payload container and a 6 degrees of freedom (DoF) platform for VTOL on harsh sea conditions [16, 17].
Agriculture	Crop spraying and monitoring with the specific pumping accessories like those found in DroneAssemble [18] and inbuilt thermal sensing camera
General Land Monitoring	Loitering services up to +10 h endurance using a high-quality camera, such as Zenmuse P1 [19]
Infrastructure Management	Slow outdoor missions for building, factory, wind turbine and potentially oil extraction site infrastructure data collection using Zenmuse P1 [19]
Large Scale Deliveries	No particular mission/payload yet constrained by our original mission requirements

3.2 What are the forces that will determine how the UAV market evolves?

Following the analysis from McKinsey&Company, growth in the drone segment will depend on:

1. Trust from audience
2. Ease of regulation
3. Advancements in technology

Reports by PwC [20] indicate that 9 out of 10 respondents believe drones can have a very positive impact in infrastructure and agriculture. However, this figure drops to 7-8 for supply delivery missions between hospitals, such as organ transplants. UAV growth in scuh segments may remain slow until public acceptance increases and regulation eases; another reason why we opted for the former markets.

Regarding advancements in technology, in both the agricultural and infrastructure sectors, advancements in drone technology will significantly influence market evolution. Currently, crop spraying and land monitoring are becoming standard practices. However, future applications such as livestock handling and crop seeding are expected to expand this market further, requiring greater technological capabilities. For infrastructure, such advancement will enable UAV missions to play a critical role in various areas, including surveillance support during fire incidents, situational awareness and post-disaster mapping and inspection of built infrastructure objects such as bridges and roads. While many of the missions carried out by the markets selected may be undertaken by smaller drones, more complex and longer-duration missions are better suited for Fixed-wing VTOL UAV.

Having understood the growth drivers of the drone market, it is evident that our drone will positively impact the market and fill the gap for a versatile, multi-use UAV.

3.3 How can UAV stakeholders capture or create value?

The following are our key statements supporting why stakeholders should work with HALO and why investors should fund our venture:

1. **Revolutionary Product Development:** Our UAV is designed to outperform current industry standards by significantly extending flight endurance and payload capacity and type at a competitive price point. For a more extensive cost analysis one can refer to section 8.
2. **Hybrid Technology and Sustainability:** By integrating hybrid propulsion technology, and potentially Sustainable Aviation Fuel (SAF) instead of Jet-A in the near future, we are committed to reducing carbon emissions and promoting environmental sustainability within the UAV industry. This approach aligns with global efforts towards greener technologies and meets increasing regulatory demands for sustainable aviation [21].
3. **Enhancing Sustainability Across Industries:** Our missions are designed to enhance sustainability across various sectors by reducing operational costs, improving worker safety, and promoting automation. This is particularly critical in addressing labour shortages in developed countries and optimising resource management in agriculture.
4. **Disaster Relief and Modular Capabilities:** We are pioneers in developing a disaster relief drone equipped with modular add-ons. This design feature caters to a variety of clients: large land holders who require extensive area monitoring for crop and livestock management, insurance companies analysing infrastructure projects, and NGOs delivering disaster relief supplies to affected zones. This revenue diversification strategy is essential for ensuring the economic viability and success of our product.
5. **Competitive pricing, profitable project and early break even**: Our tight cost control and the attractive pricing strategy that stimulates demand, result in an attractive project with operating margins above 40% from year 5 onwards. Refer to Section 8 for further detail.

For these reasons, we expect to create value stakeholders such as:

- **Non-Governmental Organisations (NGOs)**: Looking for effective disaster relief and humanitarian aid delivery systems.
- **Doctors Without Borders**: Seeking efficient and reliable methods for medical supply delivery and patient care in remote or disaster-affected areas.
- **Governments**: Focused on public safety, infrastructure monitoring, avoid overrun costs, and enhancing disaster response capabilities.
- **Agricultural corporations**: Interested in improving crop monitoring, spraying efficiency, and addressing labour shortages.
- **Infrastructure companies**: Seeking solutions for timely project management, enhanced safety inspections, and cost reduction.
- **Environmental agencies**: Focusing on sustainable practices to reduce environmental impact.
- **Technology investors**: Attracted by innovative hybrid UAV technology and the potential for significant market disruption.
- **Insurance companies**: Interested in mitigating risk through enhanced safety measures and real-time data collection of infrastructures/companies.

4 Case studies

Based on our market approach, we developed 2 major cases studies on agriculture and disaster relief missions - showing the capacity of our drone to achieve advancements in these areas:

Regarding the former, agriculture, for typical spraying rates of 8 L/min and 150 L/ha of pesticide, a mission with a 50 kg payload would take approximately 6 minutes and use 120 g of fuel.

Figure 4. Traditional spraying mechanism [22]

With a fuel capacity of 6.5 kg, as specified by FATT04 [23], REACH would perform up to 50 missions before refuelling even at 90% fuel capacity. Battery recharging would be required however every 3 to 4 missions, taking 10 to 15 minutes extra[23]. In 50 missions, the UAV could cover 16.67 ha using 2500 L of pesticide in just 6 to 8 hours. This method is more fuel-efficient, avoids crop damage from truck wheels like those in Fig. 4, and reduces pesticide usage from precise spraying techniques, making our UAV an efficient eco-friendly solution.

For the latter, disaster relief, the 2011 Great East Japan Earthquake was taken as a reference, when a 9.1 Richter scale magnitude earthquake hit the region of Tohoku. The affected area experienced communication outages for 28 days, with laptops, mobile phones, and landlines being the most impacted [24] and up to 20.000 people went missing or dead. Ensuring effective communication during the acute phase of large disasters is essential for better management of the affected zones and avoiding casualties.

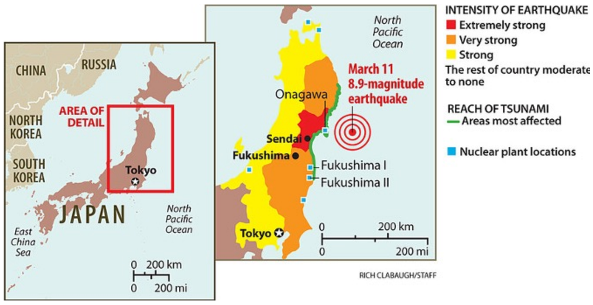

Figure 5. Affected zone map [25]

A portable, battery-powered satellite communication device, capable of transmitting substantial data volumes reliably is said to be the solution to this problem [24]. REACH's communication hub can cover an area of 19.6 km^2 from its position [26]. In this scenario, the UAV could have been deployed from the opposite coast of Japan (150 km from Tohoku's coast), ensuring the safety of the ship and assisting the disaster medical assistance teams (DMATs) in managing the area more effectively - and saving more lives by prioritising the most affected regions. Individuals could have reached out to their relatives and emergency authorities, offering a sense of calm and control. Overall, REACH would have assisted DMATs and local authorities in managing the disaster more effectively, leading to reduced casualties and greater tranquillity among the general population in such desperate conditions.

5 Branding - Who are we?

To provide a comprehensive understanding of how our business will operate, we have analysed the potential life cycle of the product and the marketing strategy behind it. These analyses can be found in Section 6. First, however, we will delve into who we are - what is HALO REACH and how our branding reflects disaster relief as our primary mission. The team collectively brainstormed and selected our company name in a session detailed further in my coordinator's report [2]. We finalised on HALO as the company name, Humanitarian Aid Logistics Operations, and REACH, the UAV name, Rapid Emergency Aid and Communications Hub.

(a) "H" finalist **(b)** "UAV fancy" finalist **(c)** "UAV simple" finalist

Figure 6. Last three design choices for the logo

Following the naming process, I began developing our logo. HALO REACH aims to address ESG concerns with a strong connection to natural landscapes, using green, blue, and beige colours symbolising the earth's pristine state. The logo design focused on simplicity and clarity, crucial attributes identified in market analysis as lacking in current UAV branding. After selecting the colour palette, I explored several logo prototypes, including designs featuring UAVs and variations with the letter "H" or brand elements. The team reviewed and narrowed down to the final three choices seen in Fig. 6, each representing the top selection from different categories ("H" - "UAV fancy" - "UAV simple").

A vote was conducted collaboratively, and the blue design was finally chosen. We deemed this design appropriate as blue can represent both the sea and the sky, fostering a sense of peace and calm. This aligns with the company's aim to promote a sense of tranquillity in our disaster relief efforts.

6 Life cycle and marketing strategy

Understanding the life cycle of HALO REACH was critical for our success. Being this drone a new product, we asked ourselves how and to what extent the shape and duration of the life cycle could be predicted - as Harvard Business prompts doing [27]. We searched for current market trends and advisory reports to predict the product life cycle and define a marketing strategy accordingly.

We based our conclusions on two reports: one on the Australian and Global UAV market [28], which highlighted the years needed for different UAV missions to reach maturity, and another report on eVTOLs life cycle in the UK [29]. These reports indicated that agriculture and infrastructure UAVs take 5-10 years to reach maturity and that disaster relief and safety operations UAVs over 10 years [28, 29]. The life cycle seen in Fig. 7 was adjusted accordingly in parallel with the financial analysis, impacting the company's business strategy.

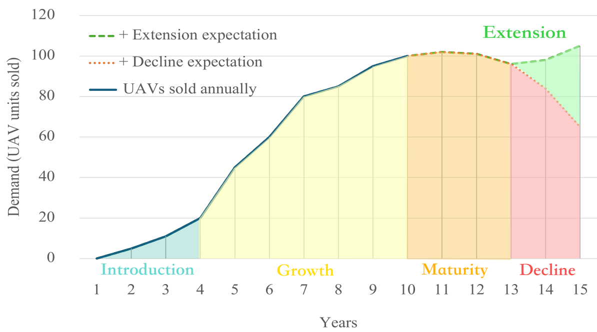

Figure 7. Life cycle of UAV

6.1 Introduction Stage (2024-2027)

Our efforts during the introduction stage centred on identifying the specific needs in disaster relief, agriculture, and infrastructure markets. Collaborating with clients such as farmers, disaster management teams and infrastructure companies will be key to understanding their needs better and to directing our selling efforts.

The company will develop UAV prototypes with a focus on key functionalities for stakeholders and potential clients during this phase. Feedback and extensive testing will be performed in controlled environments to ensure the safety, reliability, and performance of the UAV prototypes. Field trials will be conducted in each target area (disaster relief, agriculture, infrastructure) to validate the UAV's effectiveness in real-world scenarios. Ensuring compliance with aviation regulations and obtaining necessary certifications will be critical at this stage, as explained in my co-coordinator's report [2].

During this phase we will initiate our marketing strategies to educate and attract potential customers to the benefits of REACH in disaster relief, agriculture and infrastructure. These include case studies and demonstrations to showcase the UAV's capabilities and generate interest among stakeholders. We will create an online presence through an interactive website and will interact with highly influential companies/profiles to act as ambassadors of HALO REACH. Once the company secures its first customers, the focus will shift to gathering feedback to refine and improve REACH, particularly modular add-ons and software enhancements that do not require changes to the overall UAV assembly. Securing enough funding from both equity and debt during this phase is of paramount importance. The financial analysis in Section 8 estimates our funding needs at $18 million (M). The risk profile of an industrial-technology start up such as HALO requires a conservative capital structure which we set at 2/3 equity and 1/3 debt. We try to avoid the high-yield market as well as excessively diversification among shareholders, and prioritise keeping control of the startup.

6.2 Growth Stage (2027 - 2033)

During the growth phase our focus will shift towards ensuring an approximately 80% efficiency each year in the manufacturing processes. This should allows us to meet our market objectives and achieve break-even in the 6th year as per the financial analysis. As such, we will propose various financing alternatives to our clients including flexible instalment plans, leasing options, and potential partnerships with financial institutions to offer low-interest loans. Furthermore, we will continue to decrease the cost of our UAVs annually for our clients, as illustrated in Section 8, ensuring a competitive edge among potential clients and maintaining customer retention.

Our distribution network will be strengthened to ensure timely delivery and support services. HALO will invest in a sales team working with distributors and dealers widening our market reach. Adding new features or improvements, such as better sensors, extended battery life, and enhanced software will give us a technology edge and keep us ahead of competitors. Ensuring superior customer support and after-sales services will enhance customer loyalty and maintain superior branding.

6.3 Maturity Stage (2033 - 2037)

As the market becomes saturated and reaches maturity, the key focus will shift towards retaining customers and differentiating HALO REACH from the competition better. To fulfil this objective, the sales&marketing team will design comprehensive loyalty programmes including exclusive discounts, early access to new features, and dedicated customer support. Continuous engagement with clients will include communicating the latest HALO REACH technology breakthroughs, corporate achievements and the incorporation of stakeholders and ambassadors, contributing to maintaining client interest.

The manufacturing and research teams must also be wary of competitors lowering prices and using HALO REACH's innovations as their own baseline. These teams need to prepare for price competition by optimising production costs and exploring new revenue streams, exploring new markets for UAVs beyond the initial focus areas. That will also contribute to sustain business growth and extend the product's life cycle. We could improve our value-added services with predictive maintenance, advanced UAV analytics, and integration with other technologies. Adapting UAVs for tailored solutions to meet specific client needs can also enhance customer satisfaction and retention. These are all contributions from teams in operations, not just in marketing and sales.

6.4 Extension of Product (2037 - Future)

HALO needs to anticipate this phase of the life cycle by monitoring market changes and customer/in-house demand. We will need to adjust production levels to avoid overcapacity and reduce inventory costs. The company would also consider strategic partnerships or mergers and acquisitions (M&A), but also exiting, moving to emerging markets or finding new applications for the product.

7 Meet the team

This section outlines the personnel required to ensure a collaborative environment and the successful launch of the UAV implementing all the strategies described in the sections above. First, typical hierarchical roles like COO, CTO, and CMO were considered, however, the company aims to promote well-being and work quality with a flat organisational structure as its been shown to promote communication, innovation and productivity levels [30]. This does impose, however, a well thought business strategy and thus a competent CEO to mitigate organisational risks.

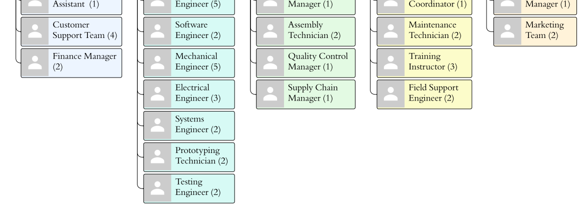

Figure 8. Team organisation chart

Fig. 8 shows the team distribution from year 1 to year 5. Initially, the R&D workforce is predominant, reflecting the focus on development in the early stages. Over time, R&D employees are expected to transition into manufacturing roles, leveraging their in-depth product knowledge into hands-on experience. This acts as a valuable asset for clients, showing clear understanding of the whole product. Administrative support remains relatively small and is not expected to grow significantly due to task automation. Conversely, the post-production, marketing, and sales teams will expand as demand increases post year 5. This structure pushes higher responsibilities but also significant potential for success, with each member playing a crucial role in the business without any redundancies.

8 Financial analysis

This section focuses on the financial aspects of the HALO REACH project, aiming to quantify its financing needs and profitability in order to attract support from lenders and equity investors during the introduction phase. The financial analysis is structured into three main sections:

1. Budgeting the direct manufacturing costs of UAVs and major expense categories to construct a basic profit&loss (P&L) account.
2. Budgeting the necessary equipment investments for UAV manufacturing.
3. Conducting cash flow and break-even analyses to estimate financing requirements.

Based on our previous market research, we preliminarily evaluated two potential business models:

1. Becoming an assembly and manufacturing company which would involve HALO manufacturing most components in-house.
2. Becoming an assembly-only company where HALO would rely on Tier-1 suppliers for the production of parts; the model followed by most manufacturers in the automobile sector.

We concluded that, during the conceptual years, HALO REACH needed to generate technological and industrial "know how" through in-house UAV manufacturing and establishing partnerships with suppliers. This strategic development of "know-how" positions HALO to change its business model effectively as it scales in the future.

Consequently, I identified the major cost categories based on standard cost structures [31, 32], upon which we specified sub categories based on our teams needs and financial management practices. Some of these categories and subcategories are shown in Fig. 9. Our analysis identified the cost of materials and components of our UAVs, requested by our manufacturing, operations and R&D teams. Additionally, we incorporated the costs associated with product characteristics for specific markets such as cameras, sensors and crop spraying mechanisms.

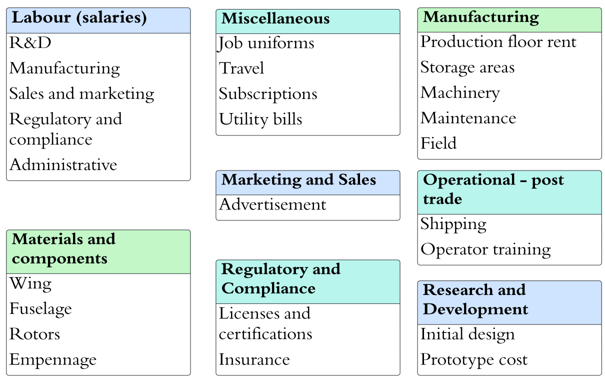

Figure 9. Cost categories and sub categories

We conducted a brainstorming session among FATT01 members to identify key structural elements of our UAVs and then consulted with other FATT teams to ensure comprehensive coverage of materials and parts. Subsequently, I researched various cost sources to estimate relevant expenses. Material costs were calculated based on manufacturers' sites [33] and the quantity required per drone was estimated by FATT03 members. Specific components' costs were based on quotes FATT01 obtained. Number of employees were established per team and salary levels used average figures for each role from survey companies like Indeed [34]. Factory and office space requirements and costs were calculated with Strategy Hat's office budgeting tools [35], resulting in a 309.74 m^2 office facility, including a 12% circulation space and amenities. The factory and office space rents were estimated at $150,000 and $29,450 respectively. Utility bills were calculated using industry data [36] and verified against Bolueta, a metallurgy company of similar size [37]. Regulatory and compliance costs were based on CAA regulations [38]. Miscellaneous expenses were estimated by comparing similar businesses and accounting for expected travel costs for the entire team amongst other cost sources. A final 15% contingency cost was added to the budget, as specified in AACE for preliminary projects [39].

Once all expenses were budgeted for year one and for one drone, the annual costs were subsequently scaled accordingly for the remaining of the forecast period. The costs for the prototype and design were only accounted for in the first two years. Salaries were adjusted for inflation using an annual CPI of 4% [40]. Estimated material costs for one UAV were multiplied by the number of drones manufactured. Manufacturing expenses related to facilities and repairs increased by 2% as we focused on efficiency. Similarly, other expenses grew less than inflation due to expected cost efficiencies.

Based on the initial budgeting exercise outlining broad expense categories, we proceeded to the next phase where we budgeted for equipment investments necessary at the project's inception. This analysis assumes that these investments are fully funded upfront and does not account for potential individual funding agreements with equipment manufacturers. We have depreciated this equipment over a period of 10 years in our profitability analysis, aligning with standard accounting practices.

Once annual expenses and investment requirements were budgeted, our focus shifted to estimating UAV production levels and demand. In the initial three years, factory production at 20-30% efficiency yielded 22 drones annually. This was the baseline for estimating manufacturing and production costs, which were subsequently scaled up based on our market analysis. Our demand analysis integrated the product life cycle considerations and drone costs to forecast annual UAV sales. Market insights and demand estimates were validated using an 80% learning curve, as shown in Fig 10a, aligning with industry norms [41]. Our estimates followed the trend on the first operating years and plateaued as maturity was reached and efficiency improvement levels plateaued too. HALO´s pricing strategy was established according to total expenses with a 35% markup, guided by industry benchmarks such as CSIMarket and companies like Vertical Aerospace [42]. It's important to note that these prices do not include VAT. For the detailed numerical analysis, please refer to Appendix D.

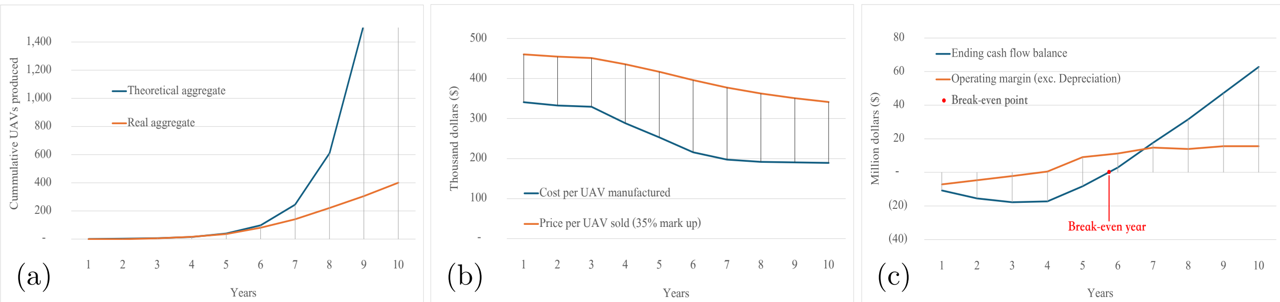

Figure 10. Financial analysis over time showing: (a) Aggregate UAV production comparison between estimate and theory [41], (b) Production and consumer UAV unit price, (c) Profit and cash statements

We also compared our costs and prices with those of our competitors to assess competitiveness. Our price and cost per UAV without the addition of modular add-ons over the projected decade illustrated in Fig. 10b, results in a competitive price of $440,000 compared to others like the SMD800 and CW-80E whose UAV prices are $465,000 and $535,000, respectively. We should emphasise that our prices go down every single year as we pass our cost efficiencies to our clients. The annual UAV price was determined by averaging the production cost per UAV from year 1 to the present year and applying a 35% margin, ensuring gradual adjustments in client prices and higher returns for the company. Even when our competitors' UAVs offer similar endurance (10/8 hours), payload capacities (50/20 kg), and slightly different GTOW (100/230 kg), HALO's competitive pricing, coupled with our emphasis on ESG and flexibility of our modular add-on options, presents a compelling value proposition.

We then generated a P&L, cash flow and break even analysis in order to quantify our financing needs and present it to investors. Our profitability analysis will stop at the operating margin level, showing an average of 40-45% margin from year 5 to year 10 by comparing our yearly operating margin to our revenues. Our operating margin and cash flow statements are shown in Fig. 10c and display the usual J curve, where initial negative cash balances precede a steep rise in returns during the growth phase [43]. Our minimum cumulative cash level determines the financing needs of HALO REACH and the moment our cumulative cash flow becomes positive establishes our break even point. These results show that we need to obtain funds for $18M, from both debt and equity investors, and that our break-even happens during year 5, during the growth phase. This aligns with other UAV companies like Cyberhawk, which broke even after 8 years in 2018. As the industry for HALO REACH is expanding, it is reasonable to expect a shorter break even point in year 5 [44]. From a funding size perspective, our $18M compares very positively with both Aveillant and Hummingbird Technologies that obtained approximately $15M for their projects [44], as these companies secured their funding prior to the rise in raw material prices since 2021 and the inflation increase since 2022.

Our financial analysis offers a detailed breakdown of expenses spanning from year 1 to year 10, providing insights into the proportion of costs per category and their evolution over time. This illustration can be seen in Appendix D. The analysis confirms that labour and R&D expenditures were central in the initial years, while manufacturing costs progressively became dominant in subsequent years due to quintupled UAV production. In summary, Table 4 presents the key points from our financial analysis.

Table 4. Financial Summary. Refer to Appendix D for the detailed numerical analysis.

Break Even	**Investment Required**	**UAV price Years 1-10**	**UAV cost Years 1-10**	**6DoF platform price**	**Zenmuse P1 price**	**Agriculture acc. price**
Year 5	$18M	$440,000 $325,000	$320,000 $180,000	$52,000 [17]	$6,700 [19]	$200 [18]

9 Conclusion

HALO is a business venture that leverages cutting-edge drone technology to support rescue missions and innovative applications in agriculture and infrastructure. Initially tasked with designing a large-scale drone for rescue missions, REACH, our market analysis revealed significant growth potential in agriculture and infrastructure, where our versatile 5-in-1 technology provides a competitive edge.

Our efficiency and demand analyses underscore our value proposition, highlighting decreasing manufacturing costs and a competitive pricing strategy. HALO requires $18 million in funding, with projections indicating an operating margin exceeding 40% after year five and a break-even point in year five. This presents an attractive investment opportunity, promising state-of-the-art technology and substantial returns.

Regarding the future, financial analyses should focus on the maturity and expansion phases of the product, ensuring accurate UAV life cycle predictions. In addition, the project will require careful management of resources and personnel to transition from an R&D-based initial stage to a manufacturing-based growth stage. This transition should incorporate the strategic decision of whether to maintain the current integrated manufacturing model or shift to an assembly-only approach.

We firmly believe this project design and scope have been successful, as the project coordination has proven effective, with all sub-teams consistently meeting project milestones and effectively addressing conflicts as they arise.

Now, HALO REACH is prepared to compete in this market.

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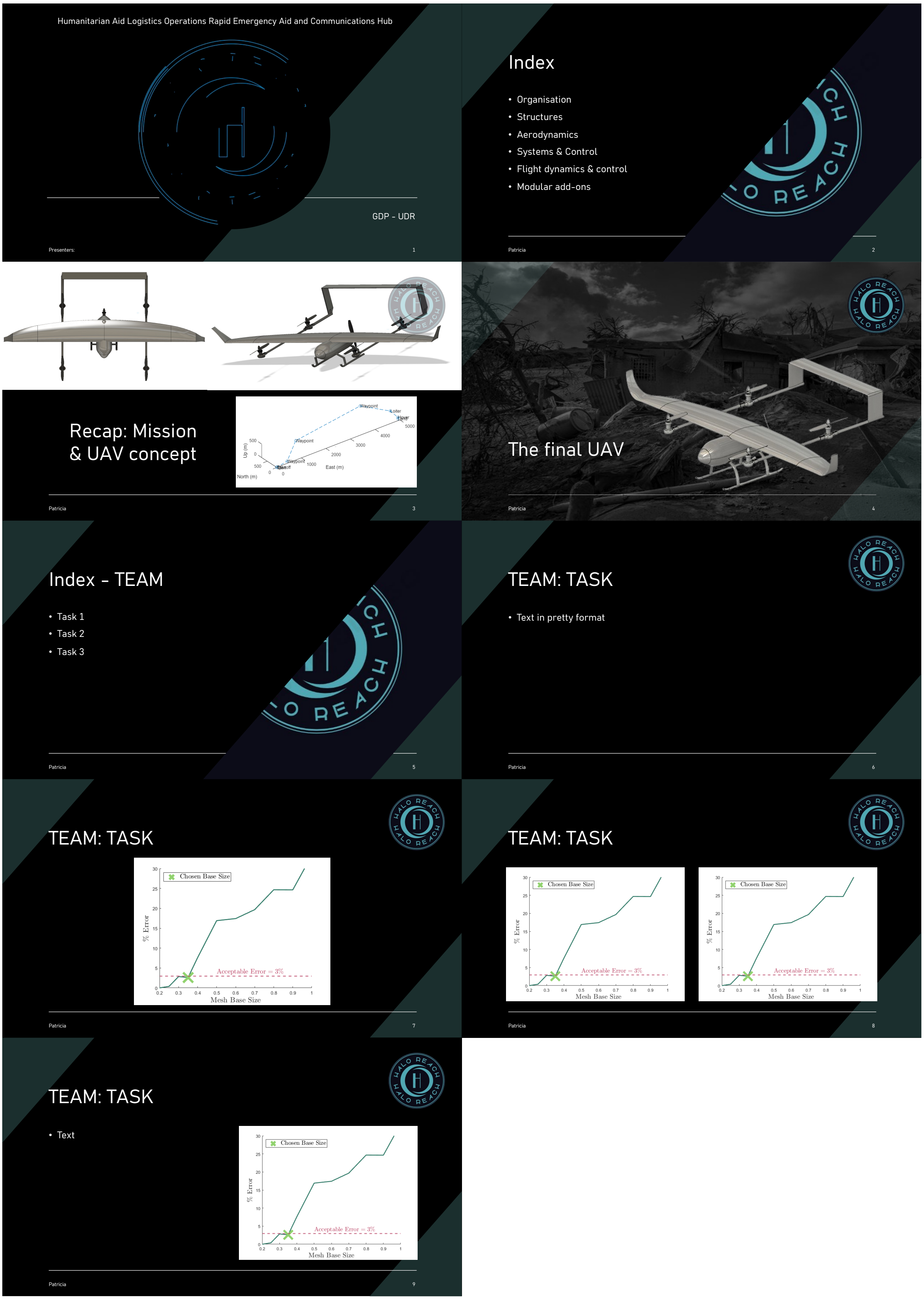
Appendix

A Weekly presentation slides' template



B Feedback forms

Below one can find the feedback form used for daily work feedback made on OneNote with its respective examples to help the users and the weekly presentation forms done via MicrosoftForms.

B.1 Daily feedback form

To write a useful daily work report one can either follow this template or use this template as a guide to write a short paragraph summarising what the team has done on a day:

- **Date and time**
- **Summary of work done and progress**
- **List of completed, ongoing, outstanding and future tasks:**
  - Completed:
  - Ongoing & outstanding:
  - Future:
- **Problems, challenges, and blockers**
- **Milestones that have been achieved**
- **Any additional information and/or useful links**
- **Key features you want to remember for tomorrow, another team and/or the next meeting with the supervisor**

Examples of this would be:

1. **Date and Time:** May 8, 2024 - 17:15
   Summary of Work Done and Progress: Today, our aviation project team made significant progress towards completing our assigned tasks. We focused on finalising the design specifications for the new aircraft wing, and researched plenty on materials and aerodynamic principles. We also reviewed... (you can enter useful links here for example)
   List of Completed, Ongoing & Outstanding, and Future Tasks:
   - **Completed:**
     - Finalised design specifications for the new aircraft wing
     - Conducted research on materials and aerodynamic principles
   - **Ongoing & Outstanding:**
     - Continuing research on alternative materials for the wing construction
     - Drafting the initial prototype design for review
   - **Future:**
     - Begin prototype construction phase
     - Schedule meetings with supervisor/subteam ”x” for design review and feedback

   Problems, Challenges, and Blockers: We encountered a setback in sourcing a specific composite material for the wing construction, but alternative options like ... are being explored.
   Milestones Achieved:
   - Finalisation of design specifications represents a significant milestone in the project timeline.

Any Additional Information: Team morale remains high, and communication channels are open for effective collaboration.
Key Features to Remember for Tomorrow/Next Meeting:

- Follow up on alternative material options for wing construction
- Prepare for upcoming meetings to review the prototype design.

2. **Date and Time:** May 8, 2024 - 17:15
Summary of Work Done and Progress: Our aviation project team made significant progress towards completing our assigned tasks. We focused on finalising the design specifications for the new aircraft wing and conducted extensive research on materials and aerodynamic principles. Additionally, we reviewed the project timeline to ensure alignment with our goals. Completed tasks include finalising design specifications and conducting research. Currently ongoing are research on alternative materials and drafting the initial prototype design for review. Future tasks include commencing the prototype construction phase and scheduling meetings with supervisor/subteam "x" for design review and feedback. We encountered a setback in sourcing a specific composite material for the wing construction, but alternative options are being explored. The finalisation of design specifications represents a significant milestone, and team morale remains high with open communication channels for effective collaboration. Key features to remember for tomorrow include following up on alternative material options for wing construction and preparing for upcoming meetings to review the prototype design.

B.2 Weekly presentation feedback form

Please go to Preview to check how the card looks to responders.

Preview

Weekly Meeting Feedback Form

Thank you for your participation. The purpose of this questionnaire is to collect your feedback, which will help us improve in the future.

* Required

* This form will record your name, please fill your name.

1

Overall, how satisfied are you with the presentation? *

Extremely dissatisfied ☆ ☆ ☆ ☆ ☆ Extremely satisfied

2

On a scale from 1-5, how informative was the presentation? *

Extremely dissatisfied ☆ ☆ ☆ ☆ ☆ Extremely satisfied

3

Could you write the sub team you are writing this feedback for? (The best thing would be to have one feedback form per sub team if possible, unless you want to do one for the general team) *

Write everyones name, you can find a doc with members' name and photo in https://imperiallondon-my.sharepoint.com/:w:/r/personal/pm220_ic_ac_uk/Documents/GDP%20Names%20and%20Photos.docx?d=w20a15a97f6924f3f810aa3ee120a08ab&csf=1&web=1&e=2nGHPg

4

How would you rate the following for the presenters? *

	Not well at all	Not very well	Somewhat well	Very well	Extremely well
Presented their outcomes, concerns and progress clearly	◯	◯	◯	◯	◯
Was concise and to-the-point	◯	◯	◯	◯	◯
Maintained my interest throughout the duration of the event	◯	◯	◯	◯	◯
Thoroughly answered questions from participants	◯	◯	◯	◯	◯

5

How satisfied are you with the following aspects of the event? *

	Very dissatisfied	Somewhat dissatisfied	Neither satisfied nor dissatisfied	Somewhat satisfied	Very satisfied
Quality of support material (figures, information, etc.)	◯	◯	◯	◯	◯
Flow of presentation	◯	◯	◯	◯	◯
Engagemente	◯	◯	◯	◯	◯
Overall content, index used	◯	◯	◯	◯	◯
Time management	◯	◯	◯	◯	◯

6

What aspects did you want to see more of in the presentation?

7

What aspects did you find useful in the presentation?

8

What objectives should we focus on more next?

9

What issues should we tackle with more priority?

10

Do you have any resource that can help us analyse .... better?

11

Would you suggest we change/reconsider anything?

12

Anything else you want to add:

C Gantt Charts

GoAERO

Imperial College

Project leads: Hasan Rashid, Patricia Moreno

Project start: **Tue, 5/14/2024**

Display week: **1**

TASK	ASSIGNED TO	PROGRESS	START	END
Preliminary Design				
Identify Sub-Teams tasks	EVERYONE	100%	5/14/24	6/11/24
Market analysis	Patricia	100%	5/14/24	5/17/24
SWOT analsysis	Hasan	100%	5/14/24	5/17/24
Produce prelimary cost estimation (sources)	Patricia	100%	5/14/24	5/17/24
Conceptual Deisgn	EVERYONE	100%	5/14/24	5/17/24
Come up with branding	EVERYONE	100%	5/16/24	5/16/24
Identify market risks	Patricia	100%	5/17/24	5/23/24
N2 diagram	Hasan	100%	5/17/24	5/24/24
AERODYNAMICS	Aerodynamics Team	100%	5/17/24	6/7/24
STRUCTURAL DESIGN AND CAD	Structures Team	100%	5/17/24	6/7/24
SYSTEMS AND PROPULSION	Systems and prop. Team	100%	5/17/24	6/7/24
FLIGHT DYNAMICS, PERFORMANCE AND CONTROL	Fligh dynamics etc. Team	100%	5/17/24	6/7/24
MODULAR ADD ON SUB-SYSTEM DESIGN	Add-ons Team	100%	5/17/24	6/7/24
House of quality	Hasan	100%	5/20/24	5/24/24
V model	Hasan	100%	5/20/24	5/24/24
Risk matrix	Patricia	100%	5/20/24	5/31/24
Life cycle analysis	Patricia	100%	5/24/24	6/7/24
Cost analysis	Project Leads	100%	5/27/24	6/7/24
Start Final presentation and pitch with current info. (trends, market gap...)	Project Leads	100%	5/27/24	6/6/24
Competitors perspective	Hasan	100%	5/31/24	6/7/24
Marketing strategy	Hasan	100%	5/31/24	6/4/24

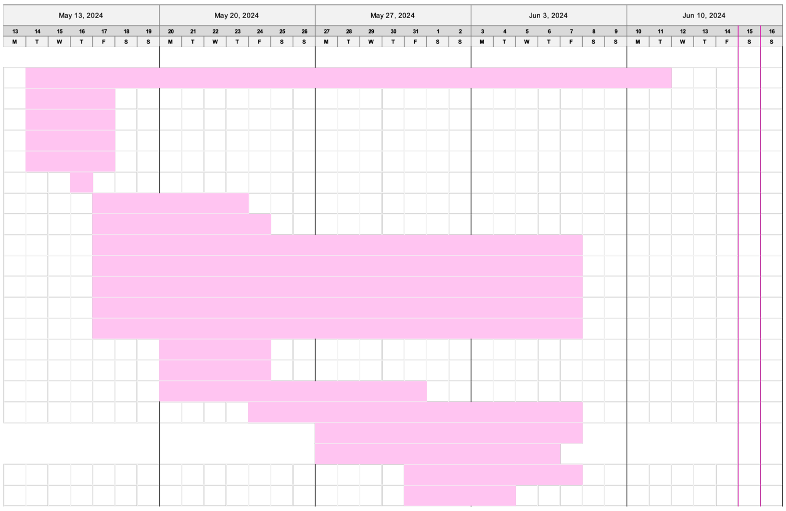


Figure 11. General Gantt Chart

GoAERO

Imperial College

Project leads: Hasan Rashid, Patricia Moreno

Project start: **Tue, 5/14/2024**

Display week: **1**

TASK	ASSIGNED TO	PROGRESS	START	END
Aerodynamics				
Rotor design: airfoil	Shahriar	100%	5/15/24	5/22/24
Wing airfoil	Max & Mat	100%	5/17/24	5/22/24
Wing lift prediction	Max & Mat	100%	5/21/24	5/25/24
Winglet design	Naboth	100%	5/21/24	5/29/24
Aerodynamic shape of where the propeller/rotors go	Aero ft. Shivani	100%	5/21/24	5/29/24
Adkins algotrithm for propeller optimisation	Parasdeep	100%	5/22/24	5/29/24
Research on pusher	Shahriar & Parasdeep	100%	5/21/24	5/30/24
CFD on wing (L/D. mesh convergence…)	Max & Mat	100%	5/22/24	5/30/24
Values for propeller	Parasdeep	100%	5/27/24	5/30/24
Geometry optimisation (XROTOR, Gblade, CFD..)	Shahriar	100%	5/27/24	5/31/24
Tail geometry aerodynamics	Naboth	100%	5/30/24	6/5/24
Airfoil optimisation w/ Adkins (BEM…)	Parasdeep	100%	Future	Future
Total aircraft drag in cruise and VTOL (CFD/VSP aero/analytical)	Team	100%	Future	Future
CFD on tail	Naboth	100%	6/3/24	6/7/24
Validation of software	Max & Mat	100%	5/31/24	6/7/24

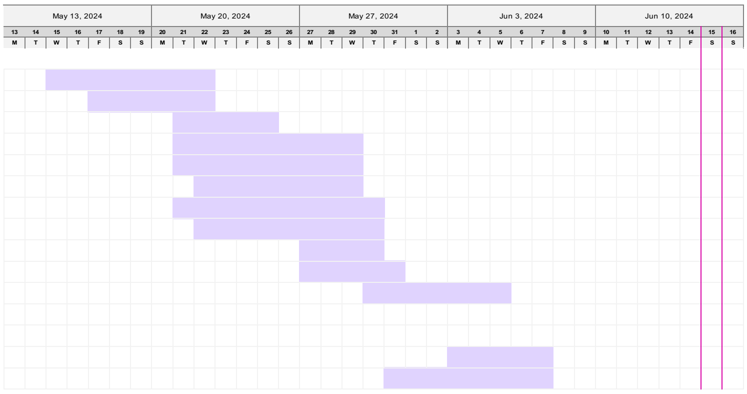


Figure 12. Aerodynamics Gantt Chart

Structural Design & CAD				
Wing Loads	Stephen	100%	5/15/24	5/23/24
Tail Loads	Aaron	100%	5/15/24	5/24/24
Boom Loads	Aaron	100%	5/15/24	5/24/24
Landing Gear Loads	Zhiyan	100%	5/15/24	5/24/24
Fuselage Loads	Shivani	100%	5/15/24	6/4/24
Rotor Loads	Stephen	100%	5/15/24	5/25/24
Material Selection	Zhiyan	100%	5/15/24	5/25/24
Material Selection	Aaron	100%	5/17/24	5/21/24
V-n	Aaron	100%	5/17/24	5/21/24
Fuselage Design	Shivani	100%	5/24/24	6/5/24
Boom Design (research on booms)	Shivani	100%	5/24/24	6/5/24
Wing design	Stephen	100%	5/24/24	6/5/24
Rotor Blade Design (loads, internal structure…)	Aaron	100%	5/24/24	6/5/24
UC Design	Zhiyan ft. Team	100%	5/24/24	6/7/24

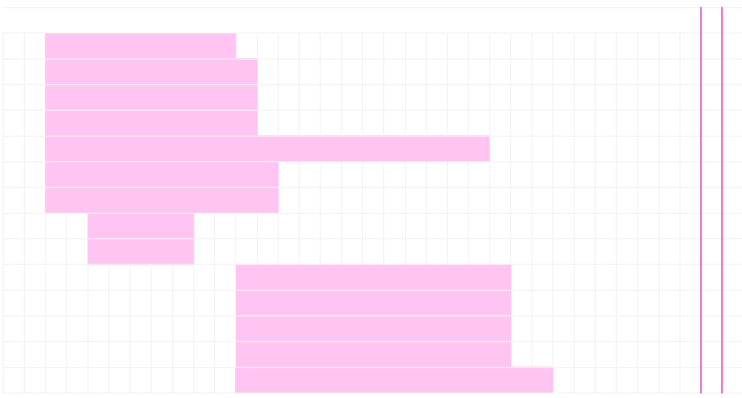


Figure 13. Structures Gantt Chart

Systems & Propulsion				
Powerplant Selection	George/Tharm	100%	5/15/24	5/24/24
Research into recharging during flight	George	100%	5/15/24	5/30/24
Specifics into takeoff and landing (batteries)	Tharm	100%	5/17/24	5/30/24
GCS + avionics	Ishaan/Yongkai	100%	5/17/24	5/30/24
Sub systems case studies	Ishaan/Yongkai	100%	5/20/24	5/24/24
Payload and add-on subsystem	Ishaan/Yongkai ft. Modular add ons	100%	5/20/24	5/24/24
System Architecture flowchart	Yongkai	100%	5/20/24	5/23/24
Communications Systems	Divij	100%	5/20/24	5/31/24
Weight and balance	Team ft. Ruairidh	100%	5/22/24	5/30/24
Physical Architecture of Control Surfaces	Ishaan/Yongkai	100%	5/23/24	5/31/24
Navigation System	Divij	100%	5/24/24	6/3/24
Fusion and processing	Ishaan	100%	5/29/24	6/5/24
Packaging and layout	Ishaan	100%	6/5/24	6/10/24
GNSS	Divij	100%	5/29/24	6/4/24
Data relay	Divij	100%	6/4/24	6/8/24
Actuators + wiring	Yongkai	100%	6/4/24	6/10/24

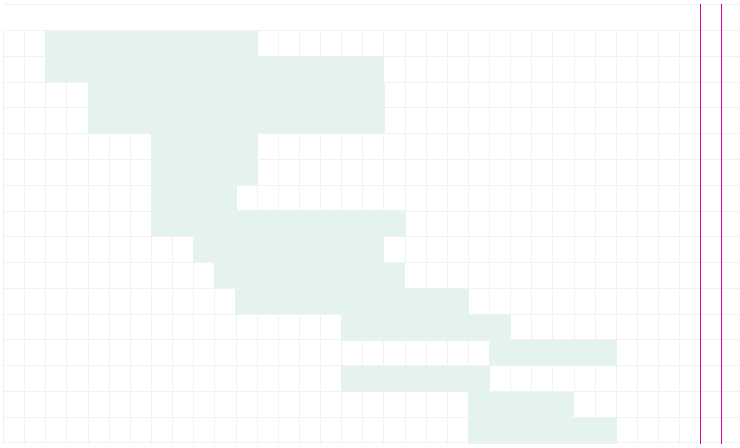


Figure 14. Systems and propulsion Gantt Chart

Flight Mechanics, Performance & Control				
Define all mission profiles	Shaun	100%	5/15/24	5/21/24
FD model of fixed	Thilak	100%	5/15/24	5/31/24
FD model of quad	Shaun/Harry	100%	5/15/24	5/31/24
First state space equations	Thilak	100%	5/17/24	5/19/24
Transitional state	Shaun/Harry	100%	5/17/24	5/31/24
Landing/Takeoff control	Harry	100%	5/19/24	6/12/24
Cruise fixed wing control	Thilak	100%	5/19/24	6/12/24
Trim Condition identification	Ruairidh	100%	5/20/24	5/27/24
Environmental Condition/Parameters estimation	Ed	100%	5/20/24	6/5/24
Simulation (Gazebo)	Ed	100%	5/20/24	6/12/24
Static Stability and Tailplane Sizing	Ruairidh	100%	5/22/24	6/2/24
Optimised path for energy	Shaun/Harry	100%	5/30/24	6/7/24

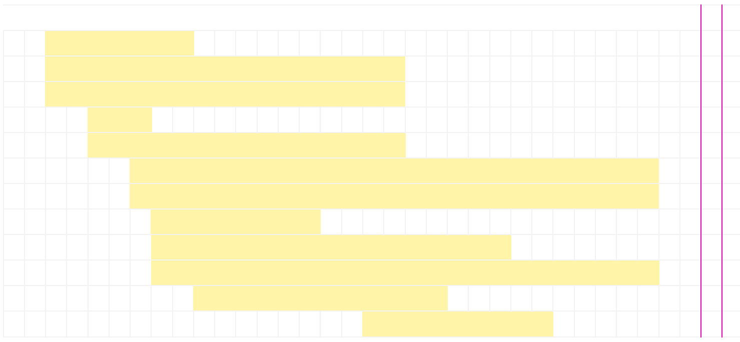


Figure 15. Flight dynamics Gantt Chart

Modular add-on sub-system design				
Research all add-on options	All	100%	5/15/24	5/21/24
Decide add ons for long endurance	Adam	100%	5/15/24	5/23/24
Decide add ons for disaster relief	Shreya	100%	5/15/24	5/23/24
Electromagnetic interference research	Shreya	100%	5/17/24	5/27/24
Decide on final placement	Adam	100%	5/17/24	5/23/24
Narrow down to final attachment	Shreya/Andrew	100%	5/17/24	5/23/24
Look at possible secondary add-on modules	Shreya	100%	5/17/24	5/28/24
Structural analysis	Shreya	100%	5/15/24	6/7/24
Aerodynamic analysis	Adam/Andrew	100%	6/3/24	6/7/24
Clamp Material Selection	Andrew	100%	5/28/24	5/30/24
Aeromechanical/vibrational analysis	Adam/Andrew	100%	6/3/24	6/7/24
Design of different payload attachments	Shreya	100%	5/30/24	5/31/24
Clamp mechanism CAD	Adam	100%	5/31/24	6/3/24
Internal Structure Design	Andrew	100%	5/30/24	6/7/24
Clamp Mechanism Design	Adam	100%	5/28/24	5/31/24
Electrical systems interaction with different payloads	Shreya	100%	5/31/24	6/7/24
Prove changeable within 10 mins	All	100%	5/31/24	6/7/24
Side-covering decision	All	100%	5/30/24	5/30/24

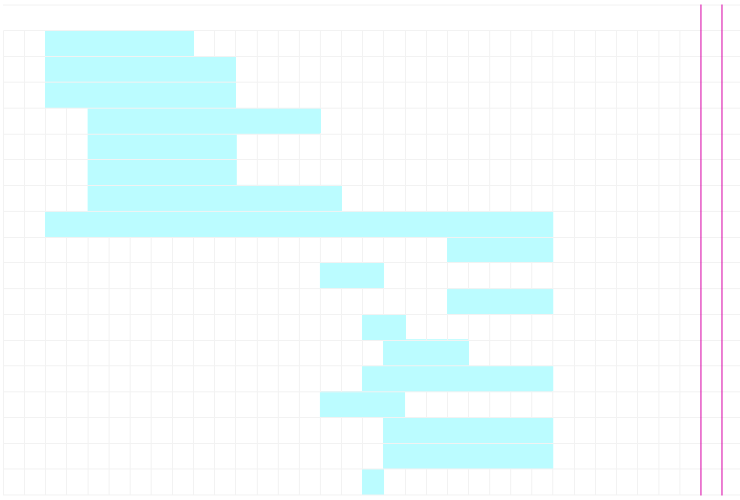


Figure 16. Modular add ons Gantt Chart

D Financial analysis

D.1 Detailed Excel sheet

HALO REACH – Costs & Investments

INVESTMENTS IN EQUIPMENT

Category	Item	Description	Amount ($)	Subtotal	%
Manufacturing	Composite Manufacturing	Composite Autoclave – ASC Econoclave	850,000.00		
		CNC Cutting Machines – HAAS VF4SS	65,000.00		
		Laminating Tables – AppliKator 4400	12,000.00		
		Vacuum Pumps – Leybold Trivac D65B	7,500.00		
		Automated Tape Layup Machine = Cincinnati VIPER 6000	1,500,000.00		
		3D Printers – Stratasys Fortus 450mc	170,000.00		
		Laser Cutting Machines – TRUMPF TruLaser 3030	200,000.00		
		Composite Moulds – custom made	30,000.00		
		Resin Transfer Moulding – Magnum Venus Products Patriot RTM System	100,000.00	2,934,500.00	79%
	Electronics and Avionics Assembly	Pick and Place Machines – Yamaha YS12	37,500.00		
		Soldering Stations – JBC CD-2BQE3	600.00		
		Reflow Ovens – Heller 1707EXL	75,000.00		
		Electronics Testing Equipment – Keysight Technologies N5183B MXG	37,500.00	150,600.00	4%
	Ancillary Equipment	Assembly Stations – custom	10,000.00		
		General Equipment – torque wrenches, screwdrivers, components	3,750.00		
		Adhesive Dispensing Systems – Nordson EFD Ultimus V	2,500.00		
		Wind Tunnels – Aerolab Educational Wind Tunnel	350,000.00		
		Flight Simulation Software – X-Plane Professional	1,500.00		
		Propulsion System Test Equipment – SF-902S Engine Dynamometer	37,500.00		
		Battery Test Equipment – Arbin BT200 Battery Tester	3,000.00		
		Structural Testing Rigs – custom	200,000.00	608,250.00	16%
		TOTAL INVESTMENTS IN		**3,693,350.00**	**100%**
		Annual depreciation of equipment (10%)		369,335.00	

Figure 17. Investment In Equipment

ANNUAL OPERATING EXPENSES ESTIMATES

			Name	Quantity (Kg. or unit)	Price ($/kg. or $/unit)	Amount ($)	Subtotal	%	Projection method
Materials and Components (per UAV)	Airframe – Wing	Upper Skin	Toray HS 3960 Prepreg Type 1	3.00	453.00	1359.00			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Lower Skin	Toray HM 3960 Prepreg Type 3	3.00	333.00	999.00			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Spars	Toray HS 3960 Prepreg Type 1	6.00	453.00	2718.00			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Stringers	Toray HS 3960 Prepreg Type 1	1.35	453.00	611.55			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Ribs	Toray HS 3960 Prepreg Type 1	3.60	453.00	1630.80	7,318.35	7%	Cost per UAV manufactured (no inflation as efficiencies assumed)
	Airframe – Fuselage	Skin	Toray HM 3960 Prepreg Type 1	2.50	453.00	1,132.50			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Heavy frames	Al2024 T861	3.50	10.05	35.18			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Light frames	Toray HM 3960 Prepreg Type 1	4.00	453.00	1,812.00			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Spars	Toray HM 3960 Prepreg Type 2	4.00	453.00	1,812.00			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Stringers	Toray HM 3960 Prepreg Type 1	4.10	453.00	1,857.30	6,648.98	6%	Cost per UAV manufactured (no inflation as efficiencies assumed)
	Airframe – Vertical Tail	Skin	Toray HM 3960 Prepreg Type 1	0.40	453.00	181.20			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Spars	Toray HM 3960 Prepreg Type 1	0.30	453.00	135.90			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Stringers	Toray HM 3960 Prepreg Type 1	0.20	453.00	90.60			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Ribs	Toray HM 3960 Prepreg Type 1	0.10	453.00	45.30	453.00	0%	Cost per UAV manufactured (no inflation as efficiencies assumed)
	Airframe – Horizontal Tail	Skin	Toray HM 3960 Prepreg Type 1	1.10	453.00	498.30			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Spars	Toray HM 3960 Prepreg Type 1	0.25	453.00	113.25			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Stringers	Toray HM 3960 Prepreg Type 1	0.30	453.00	135.90			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Ribs	Toray HM 3960 Prepreg Type 1	0.25	453.00	113.25	860.70	1%	Cost per UAV manufactured (no inflation as efficiencies assumed)
	Airframe – Booms	SKIN	Toray HM 3960 Prepreg Type 1	0.64	453.00	289.92			Cost per UAV manufactured (no inflation as efficiencies assumed)
		CORE	NOMEX Honeycomb	1.09	551.00	600.59	890.51	1%	Cost per UAV manufactured (no inflation as efficiencies assumed)
	Airframe – Rotors	Mast	Toray HM 3960 Prepreg Type 1	0.03	453.00	13.59			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Hub	Toray HM 3960 Prepreg Type 1	0.11	453.00	47.57			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Rotor blade skin	Toray HM 3960 Prepreg Type 1	0.15	453.00	67.95			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Rotor blades core	Toray HM 3960 Prepreg Type 1	0.30	453.00	135.90	265.01	0%	Cost per UAV manufactured (no inflation as efficiencies assumed)
	Airframe – Undercarriage	Strut	Toray HS 4000 Prepreg	2.60	453.00	1,177.80			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Ski	Toray HM 3960 Prepreg	2.60	453.00	1,177.80	2,355.60	2%	Cost per UAV manufactured (no inflation as efficiencies assumed)
	Modular add ons	Rack	DRF3-500 Polyacetal Rack	1.35	2.53	3.42			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Pinion	PS3-12 Nylon Pinion Gear	0.08	37.52	3.00			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Bottom plate	Epoxy/ HS Carbon fibre, UD prep	1.21	30.85	37.21			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Retractable spring reels	Aluminium	4.00	55.00	220.00			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Rod + 4 stringer design	Al 6061-T6 + Dyneema sk78 cable	4.00	11.00	44.00			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Payload container	CFRP	1.31	21.50	28.25			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Motor	MST114A293L3AA.67 (units)	1.00	60.00	60.00	395.88	0%	Cost per UAV manufactured (no inflation as efficiencies assumed)
	Propulsion	Motors:	U15 80KV (4)	4.00	689.90	2,759.60			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Pumping Mehcanism	Walbro FRB-13 Fuel Pump – Ind	1.00	337.01	337.01			
		Engine:	HIRTH 420HF	1.00	39,500.00	39,500.00			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Battery	Silicon anode	1.00	2,700.00	2,700.00	45,296.61	42%	Cost per UAV manufactured (no inflation as efficiencies assumed)
		GNSS receiver: PwrPak7 (500g)		1.00	999.00	999.00			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Pressure Barometer: DPS310 (<1g)			7.00	-			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Camera: FLiR Duo Pro R 640 (325g)		1.00	3,464.23	3,464.23			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Camera Mount: 3DR Tarot T-2D 2-Axis Brushless Gimbal Kit for IRIS+		1.00	86.99	86.99			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Sensor Processor	Nvidia Jetson AGX Xavier – 300g	1.00	1,199.00	1,199.00			Cost per UAV manufactured (no inflation as efficiencies assumed)
		Lidar	OS1 -128	1.00	11,250.00	11,250.00			
		IMU	SBG Ellipse 2 Micro Series	1.00	5,450.00	5,450.00			
		GNSS module	OEM 7700	1.00		-			
		Main and backup FCC	Pixhawk Auterion Skynode	2.00		-			
		Communication System	MILLI SAT H LW	1.00	20,000.00	20,000.00			
		Actuators and microcontrollers	TiMOTION, LINAK...	2.00	485.00	970.00			
		6 degree of freedom platform		1.00	40,000.00	-	43,419.22	40%	Cost per UAV manufactured (no inflation as efficiencies assumed)
	Sensors-Camera	Zenmuse P1		1.00	3,307.50	-	-	0%	Cost per UAV manufactured (no inflation as efficiencies assumed)
			TOTAL MATERIALS PER UAV				**107,903.85**	**100%**	

Figure 18. Annual Operating Expenses: Materials and Components (per UAV)

				Amount ($)			
Marketing and Sales	Advertising	Ads (~5% revenue)	$	100,000.00	100,000.00	3%	Grows 2% assuming improvement over inflation

			Headcount	Actual Months	$/Mth	Amount ($)			
Labour	Salaries*	R&D Team	21.0	12.0	3,179.56	801,249.00			Grows with 4% annual inflation `plus 25% adittional labour in year 5
		Manufacturing Team	5.0	12.0	3,270.13	196,207.50			Grows with 4% annual inflation `plus 25% adittional labour in year 5
		Operational-Post Production Team	8.0	12.0	2,467.60	236,890.00			Grows with 4% annual inflation `plus 25% adittional labour in year 5
		Sales and Marketing Team	8.0	12.0	2,618.56	251,381.33			Grows with 4% annual inflation `plus 25% adittional labour in year 5
		Operator Training (logistics)	3.0	12.0	2,925.00	105,300.00			Grows with 4% annual inflation `plus 25% adittional labour in year 5
		Regulatory and Compliance Team	2.0	12.0	2,967.96	71,231.00			Grows with 4% annual inflation `plus 25% adittional labour in year 5
		Administrative Support	4.0	12.0	3,409.76	163,668.67	1,825,927.50	51%	Grows with 4% annual inflation `plus 25% adittional labour in year 5
			51.0		2,976.94				

			Amount ($)			
Manufacturing	Factory Costs	Production Floor	150,000.00			Base is per UAV as of Year 1. Increasing with no. UAVs (conservative thus no inflation adjustment)
		Storage Areas	35,000.00			Base is per UAV as of Year 1. Increasing with no. UAVs (conservative thus no inflation adjustment)
	Production	Utility bills (Electrivity, water, gas...)	221,760.00			Base is per UAV as of Year 1. Increasing with no. UAVs (conservative thus no inflation adjustment)
	QC	Quality Control				
	Cleaning	Cleaning Fixed costs	59,140.00			Base is per UAV as of Year 1. Increasing with no. UAVs (conservative thus no inflation adjustment)
	Maintenace	Repairs (fixed cost)	169,001.25	634,901.25	18%	Base is per UAV as of Year 1. Increasing with no. UAVs (conservative thus no inflation adjustment)

			Amount ($)			
Research and Development	Design	Initial Design	10,000.00			Only Year 1
	Prototyping	Prototype Cost	250,000.00			Only Year 1
	Software	Development	125,000.00			Grows 2% assuming improvement over inflation
	Software	CAD/CAM software - Siemens NX (+ 26 Units @ $9552)	248,352.00			
	Testing	Flight Tests	200,000.00	833,352.00	23%	Grows 2% assuming improvement over inflation

			Amount ($)			
Operational - post trade	Shipping	Logistics	25,000.00	25,000.00	1%	Grows 2% assuming improvement over inflation

			Name	Quantity (Kg. or unit)	Price ($/kg. or $/unit)	Amount ($)			
Regulatory and Compliance	Certifications	Licenses	Units	2.0	31.13	62.26			Grows 2% assuming improvement over inflation
	Insurance	Coverage				3,500.00			Grows 2% assuming improvement over inflation
		Fixed				42,000.00			Grows 2% assuming improvement over inflation
	Legal***	Legal Fees				6,500.00			Grows 2% assuming improvement over inflation
		Renewal of Legal Fees				500.00	52,562.26	1%	Grows 2% assuming improvement over inflation

			Amount ($)			
Other operating expenses	Misc Expenses	Office supplies	7,750.00			Grows 2% assuming improvement over inflation
		Supplies Fixed	21,700.00			Grows 2% assuming improvement over inflation
		Plant emergencies	5,000.00			Grows 2% assuming improvement over inflation
		Travel (Business trips, conferences, and client meetings)	75,000.00			Grows 2% assuming improvement over inflation. Doubles from year 3
		Bank charges	1,000.00			Grows 2% assuming improvement over inflation
		Job uniforms and gear	5,000.00			Grows 2% assuming improvement over inflation
		Job search	5,000.00			Grows 2% assuming improvement over inflation
		Gifts/Donations	3,000.00			Grows 2% assuming improvement over inflation
		Subscription (market analysis, etc.)	4,000.00	127,450.00	4%	Grows 2% assuming improvement over inflation
ANNUAL OPERATING EXPENSES				3,599,193.01	100%	

Figure 19. Annual Operating Expenses: Marketing and Sales, Labour, Manufacturing, R&D, Post-trade, Regulatory and other operating expenses

In $
Inflation 4%

Operating Margin on HALO REACH

	1	2	3	4	5	6	7	8	9	10
UAVs manufactured annually	22	22	23	30	40	60	80	90	95	100
UAVs sold annually	0	5	11	20	45	60	80	85	95	100
Revenues on UAVs sold	**-**	**2,201,367**	**4,806,388**	**8,427,491**	**18,115,128**	**22,898,727**	**29,050,580**	**29,609,311**	**31,979,661**	**32,708,702**
Materials & Components on UAVs manufactured	2,373,885	2,373,885	2,481,789	3,237,115	4,316,154	6,474,231	8,632,308	9,711,346	10,250,866	10,790,385
Marketing & Sales	100,000	102,000	104,040	106,121	108,243	110,408	112,616	114,869	117,166	119,509
Labour Expenses	1,825,928	1,898,965	1,974,923	2,053,920	2,136,077	2,221,520	2,310,381	2,402,796	2,498,908	2,598,864
Manufacturing on UAVs manufactured	634,901	647,599	660,551	673,762	687,238	700,982	715,002	729,302	743,888	758,766
R&D	808,517	559,487	570,677	582,090	593,732	605,607	617,719	630,073	642,675	655,528
Operational post trade	25,000	25,500	26,010	26,530	27,061	27,602	28,154	28,717	29,291	29,877
Regulatory & Compliance	52,562	53,614	54,686	55,779	56,895	58,033	59,194	60,378	61,585	62,817
Other Operating Expenses	127,450	129,999	132,599	135,251	137,956	140,715	143,529	146,400	149,328	152,315
Depreciation of equipment	369,335	369,335	369,335	369,335	369,335	369,335	369,335	369,335	369,335	369,335
Total expenses	6,317,577	6,160,383	6,374,610	7,239,905	8,432,691	10,708,433	12,988,238	14,193,216	14,863,042	15,537,396
Overuns 15%	947,637	924,057	956,191	1,085,986	1,264,904	1,606,265	1,948,236	2,128,982	2,229,456	2,330,609
Operating Margin	**(7,265,214)**	**(4,883,073)**	**(2,524,413)**	**101,601**	**8,417,534**	**10,584,029**	**14,114,106**	**13,287,113**	**14,887,163**	**14,840,696**
	n.a.	-221.8%	-52.5%	1.2%	46.5%	46.2%	48.6%	44.9%	46.6%	45.4%
Cost per UAV manufactured	330,237	322,020	318,730	277,530	242,440	205,245	186,706	181,358	179,921	178,680
Price per UAV sold (35% mark up)	445,820	440,273	436,944	421,375	402,558	381,645	363,132	348,345	336,628	327,087

CASH FLOW ON HALO REACH

	1	2	3	4	5	6	7	8	9	10
Initial cash flow balance	-	(10,589,229)	(15,102,967)	(17,258,045)	(16,787,109)	(8,000,241)	2,953,124	17,436,565	31,093,012	46,349,510
Investments on equipment	(3,693,350)									
Operating margin (exc. Depreciation)	(6,895,879)	(4,513,738)	(2,155,078)	470,936	8,786,869	10,953,364	14,483,441	13,656,448	15,256,498	15,210,031
Ending cash flow balance	**(10,589,229)**	**(15,102,967)**	**(17,258,045)**	**(16,787,109)**	**(8,000,241)**	**2,953,124**	**17,436,565**	**31,093,012**	**46,349,510**	**61,559,541**
						Break even				
Cashflow needs for the project	**(17,258,045)**									
Breakeven point (year with first positive cash balar	**6**									
Funding sources (Equity and debt)										
Learning curve example:										
Units doubled	1	2	4	8	16	32	64	128	256	512
Application	1.00	0.80	0.64	0.51	0.41	0.33	0.26	0.21	0.17	0.13
Theoretical aggregate	1	3	6	16	39	98	244	610	1,526	3,815
Real aggregate	-	-	5	16	36	81	141	221	306	401

Figure 20. Profit and loss statement for the decade

D.2 Expense % per sector

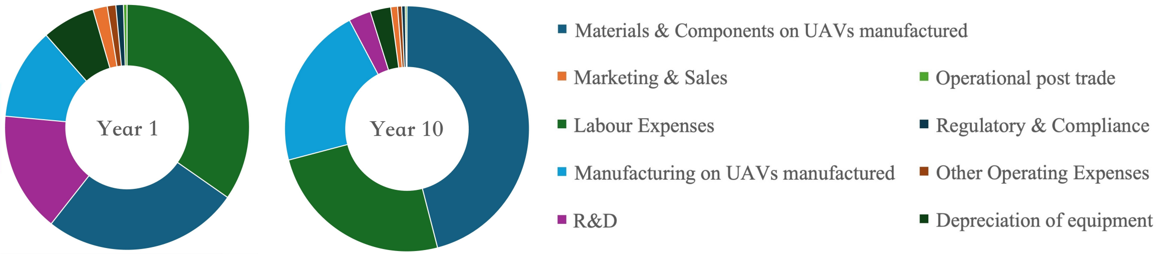


Figure 21. Cost proportion per category comparison between year 1 and year 10